University of Ottawa

Department of Mathematics and Statistics

MAT 1302C: Mathematical Methods II
Instructor: Hadi Salmasian

Midterm 1 – Group A
October 10, 2013

Surname _____ First Name _____

Student # _____ DGD _____

Instructions:

(1) You have 80 minutes to complete this exam.
(2) Show your work and justify your answers to receive full marks. Partial marks may be awarded for making sufficient progress towards a solution.
(3) All work to be considered for grading should be written in the space provided. The reverse side of pages is for scrap work. If you find that you need extra space in order to answer a particular question, you should continue on the reverse side of the page and indicate this **clearly**. Otherwise, the work written on the reverse side of pages will not be considered for marks.
provided.
(4) No notes, books, calculators or scrap paper are allowed.
(5) The final page of the exam may be used for scrap work.
(6) Good luck!

Please do not write in the table below.

Question	1	2	3	4	5	6	Total
Maximum	3	4	2	4	2	5	20
Score							

1. **(3 points)** Determine if the vector equation

$$x_1 \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} -2 \\ -4 \\ -2 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} + x_4 \begin{bmatrix} 3 \\ 0 \\ -3 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ 4 \end{bmatrix}$$

is consistent or inconsistent. If it is consistent, determine its general solution.

2. **(4 points)** Is the following linear system consistent? If so, find the general solution and write it in **vector parametric form**.

$$\begin{array}{rcrcrcrcl} x & + & 2y & - & 2z & + & 3w & = & 2 \\ 2x & + & 4y & - & 3z & + & 4w & = & 5 \\ 5x & + & 10y & - & 8z & + & 11w & = & 12 \end{array}$$

3. **(2 points)** Find <u>all</u> of the value(s) of h for which the following system has a unique solution:

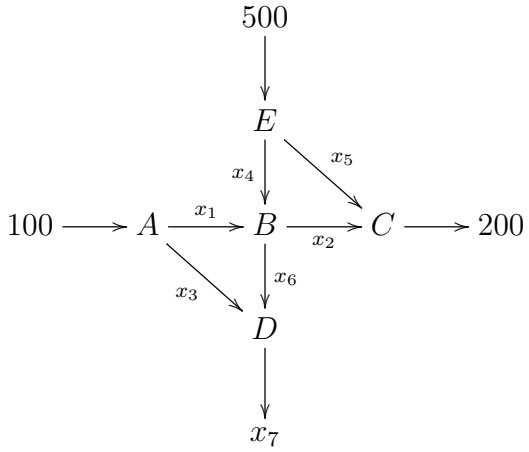
$$\begin{aligned} x + y &= 7 \\ -9x + hy &= 8 \end{aligned}$$

4. **(4 points)** Suppose that $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 2 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 2 \\ 1 \\ 1 \\ 2 \end{bmatrix}$. Is the vector $\mathbf{b} = \begin{bmatrix} 4 \\ 3 \\ 1 \\ 2 \end{bmatrix}$ in $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$? Justify your answer!

5. **(2 points)** For what value(s) of h are the following vectors linearly independent? Justify your answer!

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 3 \\ 1 \\ 3 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -1 \\ 2 \\ 4 \\ -3 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 1 \\ -1 \\ -3 \\ h \end{bmatrix}$$

6. **(5 points)** Consider the traffic flow described by the following diagram. The letters A through E label intersections and numerical values indicate flow in cars per minute. The arrows indicate the direction of flow (all roads are one-way).



(a) Write down a linear system describing the traffic flow, as well as all constraints on the variables $x_i, i = 1, \ldots, 7$. (Do not solve the linear system at this stage.)

(b) The reduced echelon form of the augmented matrix corresponding to the linear system in part (a) is:

$$\left[\begin{array}{ccccccc|c} 1 & 0 & 0 & 0 & 0 & -1 & 0 & -300 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 200 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 400 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 500 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 400 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

Find the general flow pattern.

(c) Assume that due to road work, the flow along EC is limited to a maximum of 50 cars per minute. Find the minimum flow along BC.

This page is for scrap work and is intentionally left blank.